ASTR 5140: Project 3

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1. GAIA DATA RETRIEVAL

Step 1: Using the center of the cluster, the half-light radius as a scale, and your chosen access method, download the Gaia data following the class example. As a note of caution, when you download the data, unless you have added specific conditions into the query, you will have multiple sources that do not have a measurement for at least some of its parameters (such as G_{RP}, G_{BP}, etc).

Gaia DR3 data were downloaded from the `gaiadr3.gaia_source` table using a cone search centered on NGC 362 at RA = 15.8094167° and Dec = −70.8487778°. A search radius of 0.5° (30 arcmin) was adopted. This is much larger than the cluster half-light radius of $r_h = 0.82$ arcmin, so that the downloaded field would include not only cluster stars but also the surrounding Milky Way and Small Magellanic Cloud stars needed for later membership analysis.

The query included source positions (`ra`, `dec`), astrometric quantities (`parallax`, `pmra`, `pmdec`, and their uncertainties), photometric quantities (`phot_g_mean_mag`, `phot_bp_mean_mag`, `phot_rp_mean_mag`, and `bp_rp`), and several astrometric quality indicators such as `ruwe`, `astrometric_excess_noise`, `astrometric_gof_al`, and `visibility_periods_used`. Radial velocities were also included when available, although they were not used in the main analysis.

The Gaia query result was downloaded as a `.vot` file and read into Python, then converted into a `pandas` DataFrame for easier filtering and plotting. Additional convenience columns were created, including the angular distance of each source from the cluster center in both degrees and arcminutes, as well as a manually computed color column $G_{BP} - G_{RP}$. Because many Gaia sources do not have measurements in every parameter, boolean flags were also created to track whether each source had valid color information, proper motions, and parallaxes. This made it easier to apply later cleaning cuts in a controlled and transparent way.

The final output of this step was a master Gaia table for the NGC 362 field with positional, photometric, astrometric, and quality information, together with convenience columns designed to simplify subsequent cleaning, membership selection, and dynamical analysis.

2. PHOTOMETRIC AND ASTROMETRIC DATA CUTS

Step 2: Make astrometric cuts to clean the data. For our case, we will be interested in using a Color Magnitude Diagram to investigate the data, so we will want to remove all sources that do not have a measured magnitude in either the red (G_{RP}) or blue filters (G_{BP}).

To prepare the Gaia sample for analysis, we apply a sequence of basic photometric and astrometric quality cuts to remove sources that could not be used reliably in later diagrams and membership selection. Because the color–magnitude diagram (CMD) requires both blue and red photometry, we first require all sources to have measured values of `phot_bp_mean_mag`, `phot_rp_mean_mag`, and `phot_g_mean_mag`. This ensures that each retained source has a valid color $G_{BP}-G_{RP}$ and a corresponding Gaia G magnitude.

Next, we require all retained sources to have finite values for `pmra`, `pmdec`, and `parallax`, since proper motions and parallaxes are essential for identifying likely cluster members and separating them from foreground/background contaminants. Finally, we apply a standard Gaia astrometric quality cut of RUWE < 1.4, where RUWE is the renormalized unit weight error Lindegren et al. (2021). We do not apply the more involved corrected flux-excess cut described by Riello et al. (2021).

The effect of each cut on the sample size is summarized below. Starting from the full field of 54,877 Gaia sources, requiring valid G, G_{BP}, and G_{RP} measurements reduced the sample to 44,735 stars. Requiring valid proper motions and parallaxes further reduced the sample to 38,876 stars. Applying the additional RUWE quality cut produced a final cleaned sample of 36,363 stars for further analysis. A color column $G_{BP} - G_{RP}$ was then computed for the cleaned datasets and used in the subsequent CMD analysis.

These cuts were intended only to produce a clean and usable astrometric sample. At this stage, the sample still contains cluster stars, Milky Way foreground/background stars, and stars associated with the Small Magellanic Cloud. Membership selection was deferred to later steps.

Table 1. Summary of astrometric and photometric quality cuts applied to the Gaia sample.

Sample	N_{sources}	Fraction Remaining
Raw sample	54,877	1.000
Color only	44,735	0.815
Color + astrometry	38,876	0.708
Color + astrometry + RUWE	36,363	0.663

3. INITIAL PLOTS

Step 3: Create an initial set of plots to better understand the data. A recommended first suite of plots is the following:

- **Color Magnitude Diagram** (G **vs** $(G_{BP} - G_{RP})$)
- **Proper Motion (RA) vs Proper Motion (Dec)**
- **Proper Motion (RA) or Magnitude** G **vs Parallax**
- **Position (RA) vs Position (Dec)**

Using the cleaned Gaia sample from Step 2, we construct an initial suite of diagnostic plots to visualize the photometric, astrometric, and spatial structure of the NGC 362 field. These are shown in Figure 1. The purpose of these plots is to identify the main stellar populations in the field and to determine which observables will be most useful for selecting likely cluster members.

The color–magnitude diagram (top left) shows a clear stellar sequence that is consistent with a globular cluster population, together with a broader distribution of contaminating field stars. The proper-motion diagram (top right) reveals a compact overdensity offset from $(0, 0)$ in proper-motion space, which is a strong indication of the cluster population. In contrast, a broader concentration of stars lies closer to the origin and likely includes Milky Way and Small Magellanic Cloud contamination. This immediately suggests that proper motion will be one of the most effective membership discriminants.

The parallax versus magnitude diagram (bottom left) shows that many stars in the field are concentrated near small parallaxes, as expected for a distant globular cluster and background populations such as the Small Magellanic Cloud. However, there is also a broader distribution of parallaxes that likely includes nearby Milky Way stars. This indicates that parallax can be used to suppress at least some of the foreground contamination. Finally, the sky-position diagram (bottom right) shows a strong spatial overdensity centered on the known position of NGC 362, confirming that the cluster is clearly detected spatially. The dashed circle marking the half-light radius is extremely small relative to the full 30 arcmin search region, which is expected and illustrates that the downloaded field is dominated in area by non-member stars.

Taken together, these initial plots show that the NGC 362 field contains multiple overlapping stellar populations rather than a clean cluster-only sample. In particular, the combination of a compact proper-motion overdensity, a strong spatial concentration, and a broader surrounding field suggests that cuts in proper motion, parallax, and projected radius will be effective for isolating likely cluster members in the next step.

4. ITERATIVE CUTS

Step 4: Explore a set of cuts to apply to the data to select the members of the cluster from the background Milky Way stars. This will be the most involved step of the project and may require you to iterate a few times before you will feel comfortable with your final selection of stars.

Guided by the diagnostic plots from Step 3, we apply a two-step series of iterative cuts in proper motion, parallax, and projected radius to isolate a likely member population for NGC 362. The goal of this step is not to define a final membership catalog, but rather to construct a progressively cleaner candidate sample that captures the cluster population while reducing obvious contamination from foreground and background stars.

We begin with a first-pass selection in proper-motion space by centering a circular cut on the visually identified cluster overdensity at approximately $(\mu_\alpha^*, \mu_\delta) \approx (7.0, -2.5)$ mas yr^{-1} with a radius of 3 mas yr^{-1}. We combine this with a loose parallax cut of $-0.5 < \varpi < 0.5$ mas in order to reject stars that are clearly nearby Milky Way contaminants. The resulting first-pass sample, shown in Figure 2, isolates the dominant cluster overdensity while preserving a broad range of candidate members.

We then refine the selection with a second-pass cut. In this iteration, we tighten the proper-motion selection to a circle centered at $(\mu_\alpha^*, \mu_\delta) = (6.8, -2.3)$ mas yr^{-1} with radius 2 mas yr^{-1}, retain the same parallax cut, and additionally require stars to lie within 10 arcmin of the cluster center. This radial cut is motivated by the strong spatial concentration visible in the sky-position panel.

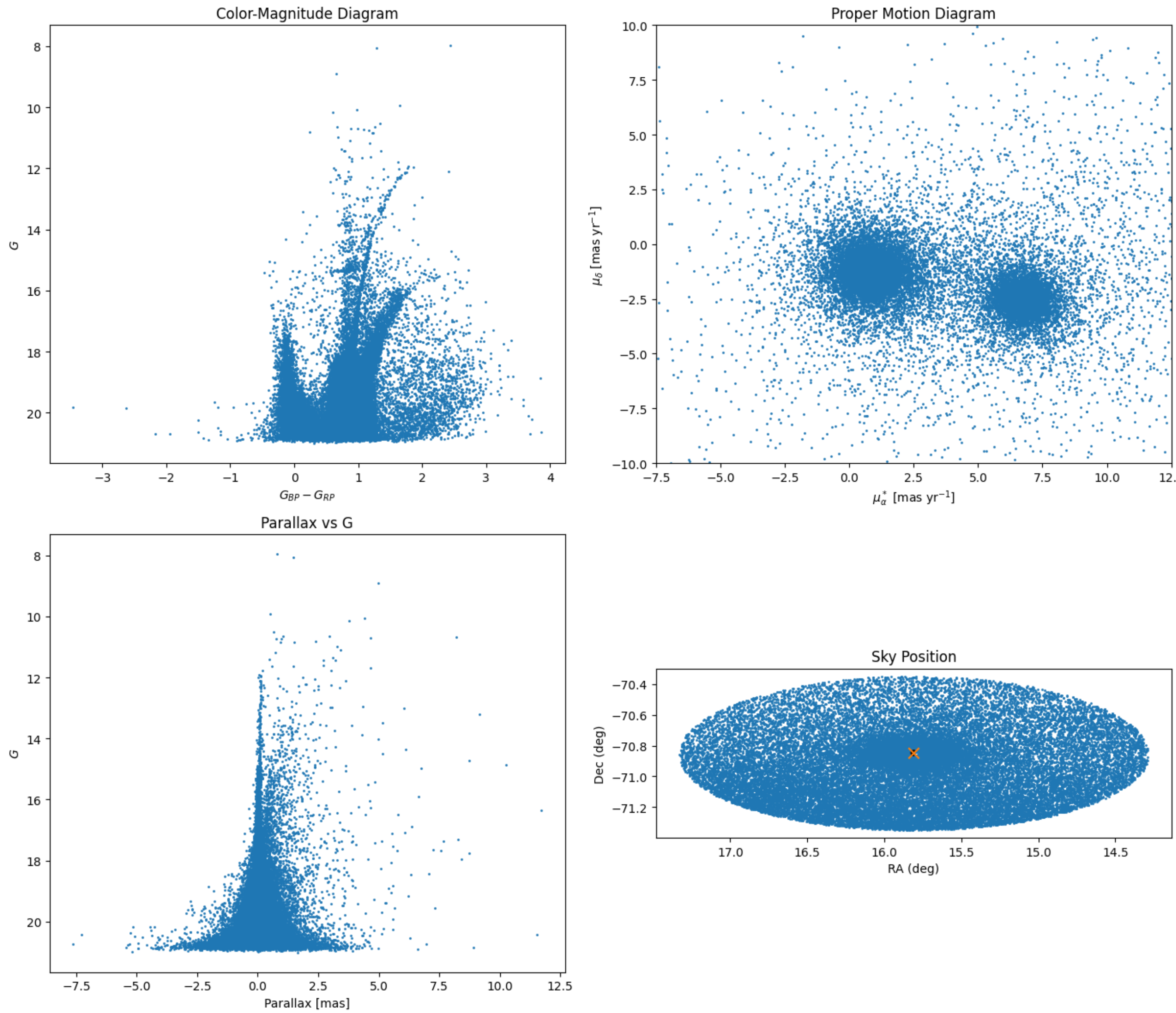

Figure 1. Initial diagnostic plots for the cleaned Gaia sample in the NGC 362 field. Top left: color–magnitude diagram. Top right: proper-motion diagram. Bottom left: parallax versus G magnitude. Bottom right: sky position relative to the cluster center, with the half-light radius shown as a dashed circle.

The resulting second-pass sample, shown in Figure 3, is substantially cleaner than the first-pass selection.

The second-pass sample produces a much more compact and coherent cluster sequence in the color–magnitude diagram, a tighter concentration in proper-motion space, and a stronger central overdensity on the sky. At the same time, the sample still shows evidence of residual contamination, particularly from populations that overlap the cluster in proper motion and parallax but differ in spatial and photometric structure. This is expected in the NGC 362 field, which contains both Milky Way stars and contamination from the Small Magellanic Cloud. For this reason, these cuts are best understood as a useful pre-selection step rather than a definitive membership classification.

Overall, the iterative cuts confirm that proper motion provides the strongest first-order separation of the cluster from the field, while parallax and projected radius significantly improve the purity of the candidate member sample. This motivates the use of a probabilistic mixture model in the next step to separate the overlapping stellar populations more rigorously.

5. GAUSSIAN MIXTURE MODEL

Step 5: Apply a Gaussian mixture model to the NGC 362 data. In contrast to the class example, the field contains multiple background popula-

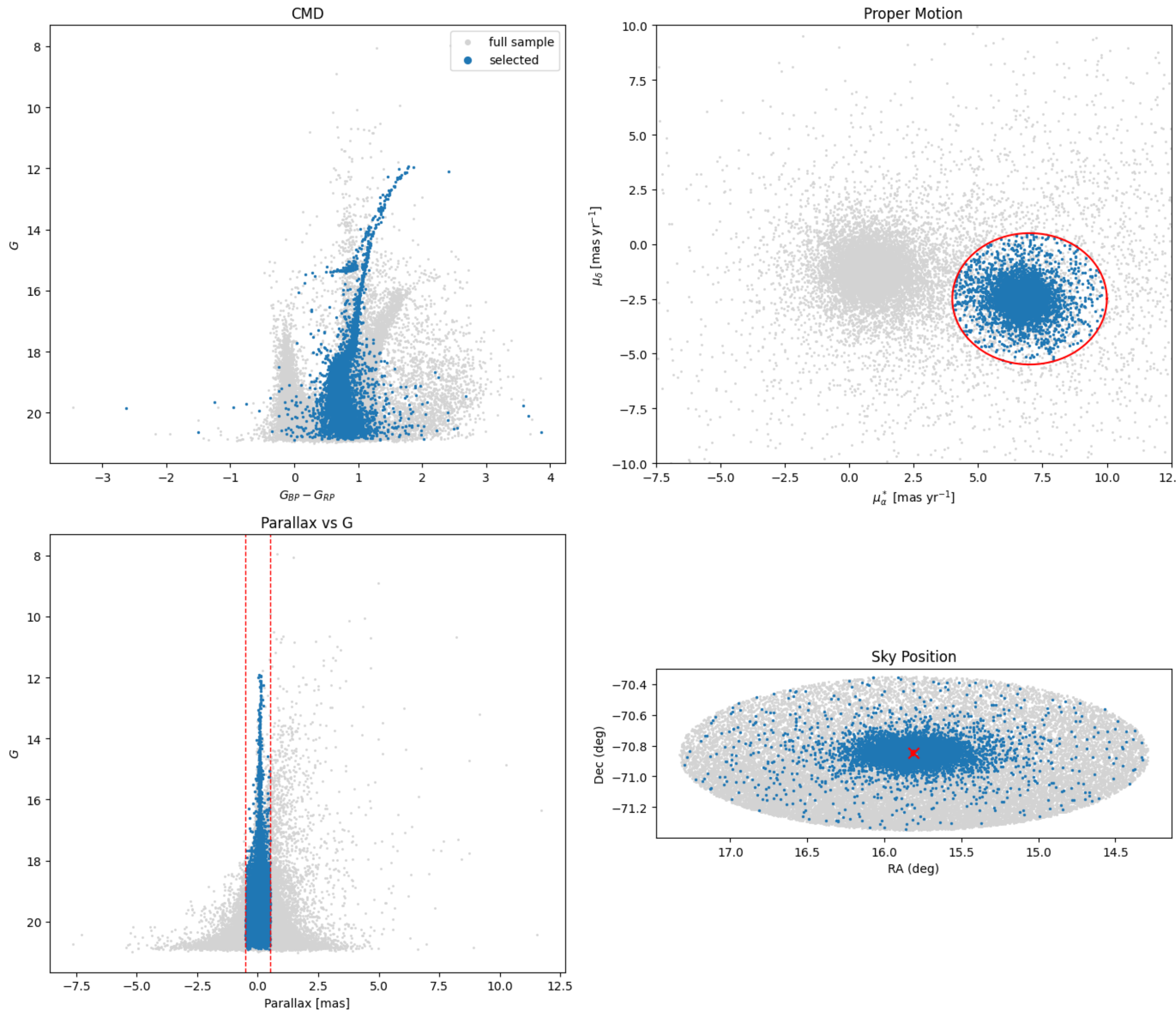

Figure 2. First-pass member selection for NGC 362. Gray points show the full cleaned Gaia sample, while blue points show stars selected using a broad proper-motion cut centered on the cluster overdensity together with a loose parallax cut. The four panels show the color–magnitude diagram, proper-motion diagram, parallax versus G, and sky position.

tions, including both Milky Way stars and the Small Magellanic Cloud.

To statistically separate the overlapping stellar populations in the NGC 362 field, we apply a Gaussian mixture model (GMM) in proper-motion space. While the iterative cuts from Step 4 isolate a relatively clean sample of candidate members, the field still contains significant contamination from both Milky Way stars and the Small Magellanic Cloud (SMC), which cannot be fully separated using hard cuts alone.

We construct the GMM using the two-dimensional proper-motion vector $(\mu_\alpha^*, \mu_\delta)$, restricting the input sample to a region of proper-motion space that contains the cluster and nearby contaminants. This reduces the influence of extreme outliers while preserving all relevant populations. The resulting sample contains $N = 32{,}923$ stars.

We first fit a two-component GMM to the data. The resulting components have mean proper motions of approximately $(0.82, -1.25)$ mas yr^{-1} and $(6.59, -2.46)$ mas yr^{-1}. As shown in Figure 4, the second component clearly corresponds to the NGC 362 cluster, while the first component represents a combination of background populations. However, the spatial and photometric distributions indicate that this background component is

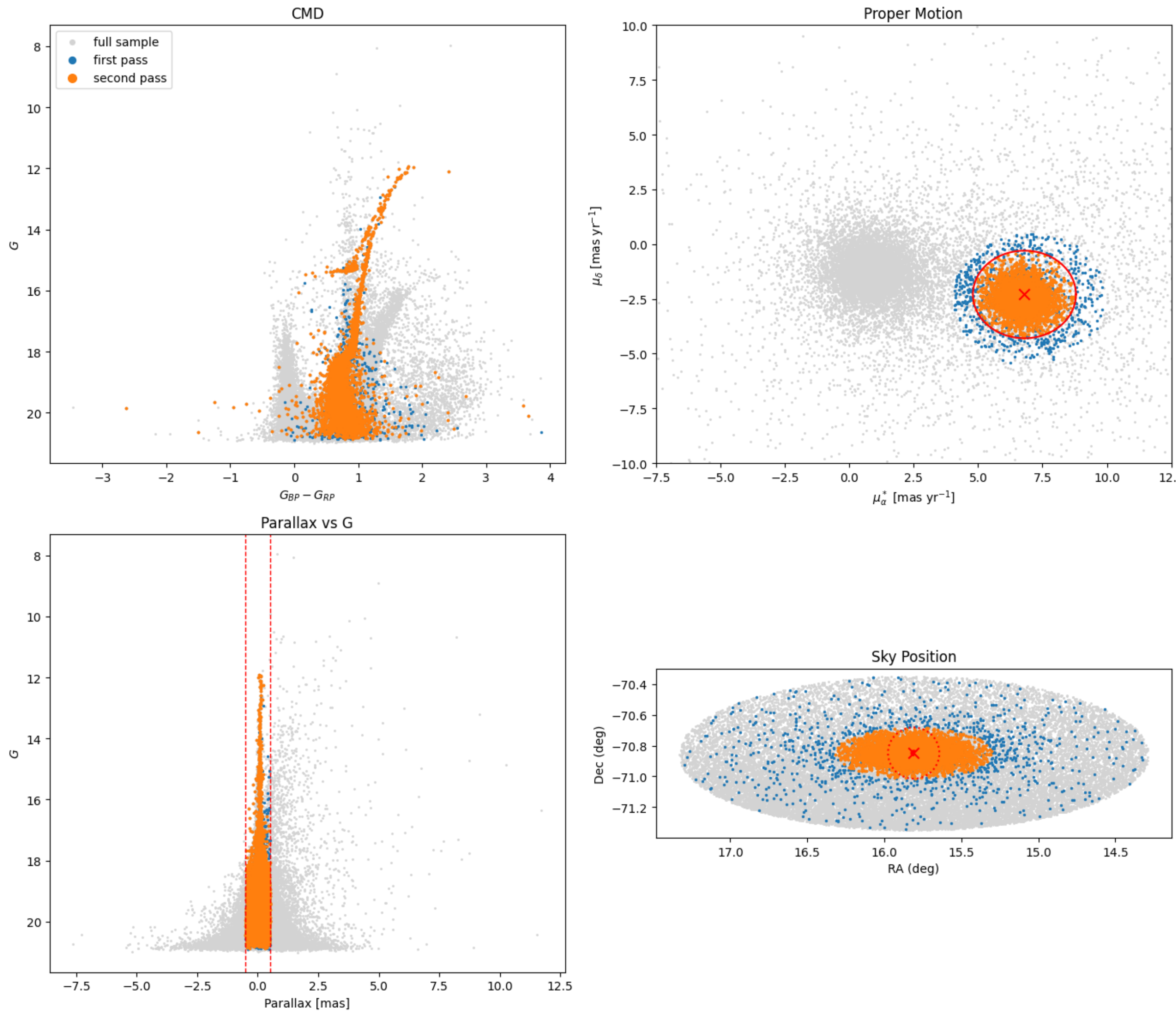

Figure 3. Second-pass member selection for NGC 362. Gray points show the full cleaned Gaia sample, blue points show the first-pass selection, and orange points show the refined second-pass sample. The tighter proper-motion cut, retained parallax cut, and additional radial cut produce a cleaner cluster candidate sample that is more centrally concentrated and more coherent in the color–magnitude diagram.

not well described by a single Gaussian, motivating the use of a more complex model.

We therefore fit a three-component GMM to account for the presence of both Milky Way and SMC populations. The three components have mean proper motions of approximately $(0.78, -1.21)$, $(6.72, -2.55)$, and $(2.94, -1.69)$ mas yr^{-1}. Inspection of the spatial and color–magnitude distributions (Figure 5) shows that:

- The component centered near $(6.7, -2.5)$ mas yr^{-1} is spatially concentrated and forms a coherent sequence in the CMD, identifying it as the NGC 362 cluster.
- The component near (0.8,-1.2) mas yr^{-1} is spatially broad and corresponds to the SMC population.
- The intermediate component near $(2.9, -1.7)$ mas yr^{-1} is more diffuse and is associated with Milky Way foreground/background stars.

Using the three-component model, we define a probabilistic cluster membership by selecting stars assigned to the cluster component with high confidence. Specifically, we require both that a star is classified as belonging to the cluster component and that its membership probability exceeds 0.8. This yields a high-confidence sample of $N = 7{,}354$ cluster members.

This probabilistic selection improves upon the hard cuts from Step 4 by accounting for overlap between populations in proper-motion space. The resulting sample is both spatially concentrated and forms a well-defined sequence in the color–magnitude diagram, indicating that the GMM successfully isolates the cluster population while reducing contamination from the Milky Way and SMC.

6. VELOCITY DISPERSION AND TIDAL RADIUS

Step 6: Using the membership list with the cuts and the Gaussian mixture model, measure the velocity dispersion in both directions and the tidal radius of the cluster.

Using the high-confidence NGC 362 member sample selected from the three-component Gaussian mixture model, we measure the proper-motion dispersion of the cluster in both astrometric directions. For the final cluster sample, we adopt only stars assigned to the cluster component with membership probability $P_{\rm cluster} > 0.8$. This yields a sample of 7,354 likely cluster members.

We compute the mean and standard deviation of the proper motions in right ascension and declination for this final membership sample. The resulting systemic proper motion and one-dimensional dispersions are

$$\langle \mu_\alpha^* \rangle = 6.7191 \text{ mas yr}^{-1}, \tag{1}$$
$$\langle \mu_\delta \rangle = -2.5564 \text{ mas yr}^{-1}, \tag{2}$$
$$\sigma_{\mu_\alpha^*} = 0.3497 \text{ mas yr}^{-1}, \tag{3}$$
$$\sigma_{\mu_\delta} = 0.3386 \text{ mas yr}^{-1}. \tag{4}$$

Assuming a distance to NGC 362 of $d = 8.6$ kpc, we convert the proper-motion dispersions into tangential velocity dispersions using

$$\sigma_v = 4.74\, \sigma_\mu\, d, \tag{5}$$

where σ_μ is in mas yr^{-1} and d is in kpc. This gives

$$\sigma_{v,\alpha} = 14.256 \text{ km s}^{-1}, \tag{6}$$
$$\sigma_{v,\delta} = 13.801 \text{ km s}^{-1}. \tag{7}$$

We adopt the mean of these two values as the characteristic projected velocity dispersion of the cluster:

$$\langle \sigma_v \rangle = 14.029 \text{ km s}^{-1}. \tag{8}$$

To estimate the tidal radius, we construct radial surface-density profiles for both the cluster and Milky Way foreground/background populations identified by the three-component Gaussian mixture model. We compute the projected angular radius of each source from the adopted cluster center using the small-angle approximation, and convert the result to arcminutes. We then bin both the cluster and Milky Way samples in annuli of width $1'$ and compute the surface density in each annulus as

$$\Sigma(r) = \frac{N(r)}{A(r)}, \tag{9}$$

where $N(r)$ is the number of stars in the annulus and $A(r)$ is its area in arcmin^2.

Rather than explicitly subtracting the background from the cluster profile, we compare the radial surface-density profile of the cluster component directly to that of the Milky Way component. At small radii, the cluster surface density is much larger than the Milky Way background, while at large radii the two become comparable. We estimate the Milky Way background level by taking the mean surface density of the Milky Way component at radii greater than $15'$ and find

$$\Sigma_{\rm MW,bg} = 2.2273 \text{ stars arcmin}^{-2}. \tag{10}$$

We adopt an empirical definition of the tidal radius as the radius where the cluster surface density first drops below the background level. Using this criterion, we find

$$r_t = 10.5'. \tag{11}$$

Finally, we convert this angular tidal radius into a physical scale using the small-angle relation

$$r_t(\text{pc}) = d\,\theta, \tag{12}$$

where θ is in radians. For $d = 8.6$ kpc, this gives

$$r_t = 26.27 \text{ pc}. \tag{13}$$

Figure 6 shows the radial surface-density profile used to estimate the tidal radius, including the Milky Way background level and the adopted crossing point.

7. VIRIAL MASS

Step 7: Use the Virial Theorem to make the final mass calculation of the cluster. Be careful of which quantities you use where.

To estimate the dynamical mass of NGC 362, we apply a simple virial-theorem scaling using the tidal radius measured in Step 6 and the characteristic projected velocity dispersion derived from the Gaia proper motions. In astrophysical units, we write the mass estimate as

$$M_{\rm vir} \approx \frac{r_t\, \sigma_v^2}{G}, \tag{14}$$

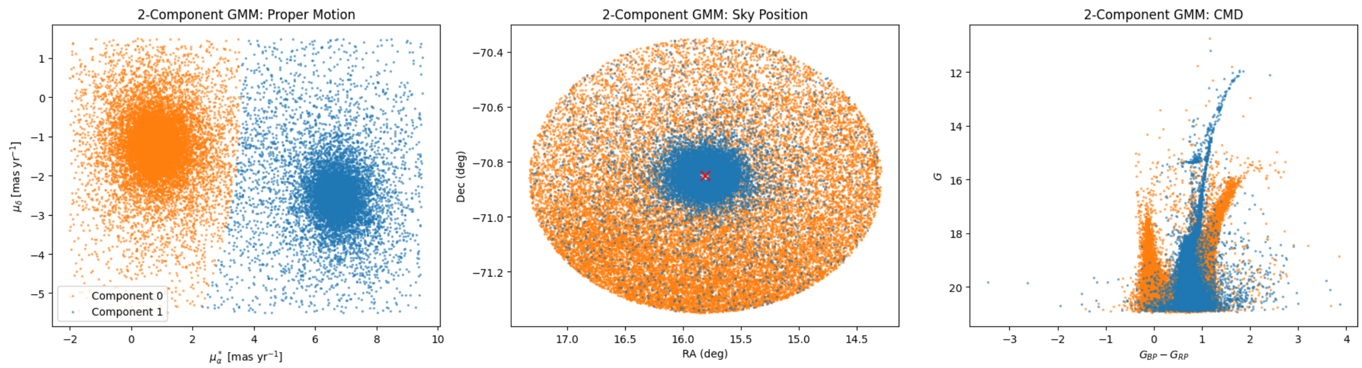

Figure 4. Two-component Gaussian mixture model applied to the proper-motion distribution of NGC 362. The model separates the cluster population from a combined background component, but does not fully resolve the distinct Milky Way and Small Magellanic Cloud populations.

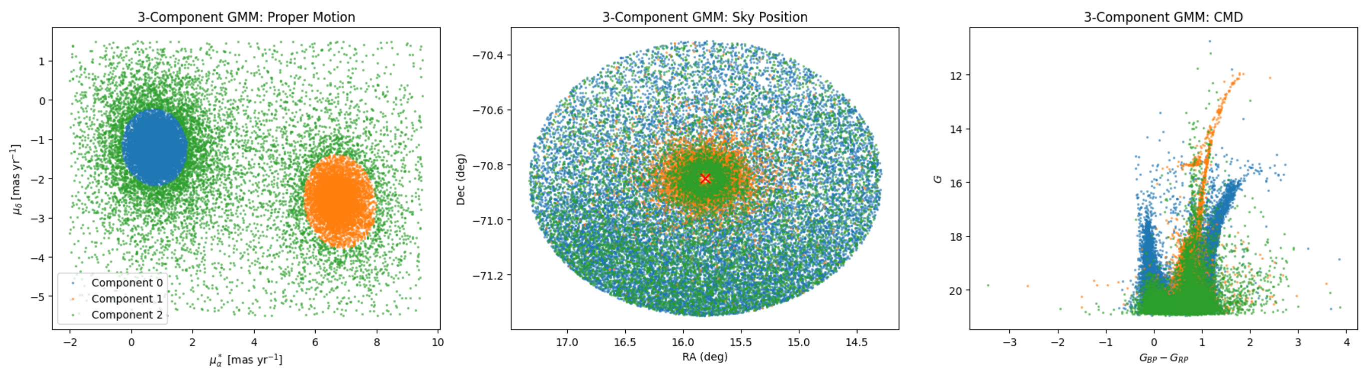

Figure 5. Three-component Gaussian mixture model applied to the proper-motion distribution. The model separates the field into three distinct populations corresponding to the NGC 362 cluster, the Small Magellanic Cloud, and Milky Way foreground/background stars. The cluster component is identified by its tight spatial concentration and coherent sequence in the color–magnitude diagram.

where r_t is the tidal radius in parsecs, σ_v is the characteristic one-dimensional velocity dispersion in km s^{-1}, and

$$G = 4.302 \times 10^{-3} \text{ pc } M_\odot^{-1} (\text{km s}^{-1})^2. \tag{15}$$

Using the tidal radius derived in Step 6,

$$r_t = 26.27 \text{ pc},$$

together with the mean projected velocity dispersion,

$$\langle \sigma_v \rangle = 14.029 \text{ km s}^{-1},$$

we obtain a virial mass of

$$M_{\text{vir}} = 1.202 \times 10^6 \ M_\odot. \tag{16}$$

As a consistency check, we also compute the virial mass separately using the tangential velocity dispersions in right ascension and declination. This gives

$$M_{\text{vir},\alpha} = 1.241 \times 10^6 \ M_\odot, \tag{17}$$
$$M_{\text{vir},\delta} = 1.163 \times 10^6 \ M_\odot. \tag{18}$$

These values agree to within a few percent, indicating that the mass estimate is not strongly sensitive to which proper-motion component is used. We therefore adopt the value based on the mean projected velocity dispersion as the final virial mass estimate for NGC 362.

This mass estimate should be interpreted as approximate. The virial coefficient depends on the detailed internal structure of the cluster, and the tidal radius is only an empirical estimate of the cluster size rather than a formal virial radius. In addition, we note that this velocity dispersion is based on projected tangential motions and therefore represents a lower-dimensional estimate of the full three-dimensional dispersion. Nevertheless, the resulting mass is physically reasonable for a massive Galactic globular cluster and provides a useful first-order dynamical estimate.

8. BINNED VELOCITY DISPERSION

Step 8: Explore the binned velocity dispersion profile (velocity dispersion versus radius). Does

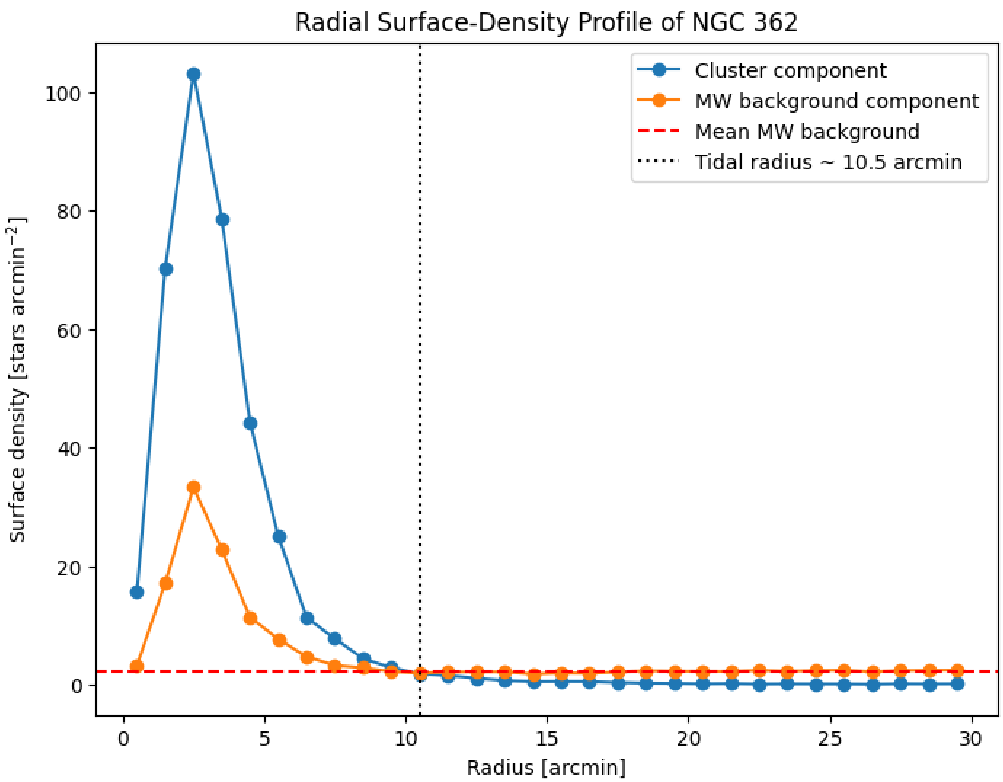

Figure 6. Radial surface-density profiles of the NGC 362 cluster component and Milky Way background component derived from the three-component Gaussian mixture model. The dashed horizontal line marks the mean Milky Way background density, and the dotted vertical line marks the adopted tidal radius at $r_t \approx 10.5'$.

this allow you to improve the mass estimate? Explain in two paragraphs.

To examine whether the internal kinematics of NGC 362 vary with projected radius, we construct a binned velocity-dispersion profile using the high-confidence GMM-selected cluster members. We divide the sample into concentric annuli of width $2'$ and measure the proper-motion dispersion separately in μ_α^* and μ_δ within each bin. These dispersions are then converted into tangential velocity dispersions using the adopted cluster distance of 8.6 kpc. The resulting radial bin centers are $1'$, $3'$, $5'$, $7'$, $9'$, $11'$, and $13'$, containing 636, 3009, 1909, 751, 371, 204, and 127 stars, respectively. Figures 7 and 8 show the resulting velocity-dispersion profile, both separated by coordinate direction and with the mean dispersion plotted together with simple statistical error bars.

The binned profile is broadly consistent with a nearly flat projected velocity dispersion, with values generally between about 12.5 and 14.5 km s^{-1} across the measured radial range. There is a mild decline toward larger radii, together with some scatter in the outer bins, but the uncertainties increase substantially where the number of stars becomes small. In particular, the slight rise in the outermost bin is not clearly significant once the error bars are taken into account and is likely driven by small-number statistics or residual contamination. Overall, the profile does not show strong evidence that the global dispersion used in Step 7 is severely biased. Therefore, while the binned profile provides a useful consistency check, it does not justify a revision of the virial mass estimate derived in Step 7. A more sophisticated dynamical analysis would fit the full radial dependence of both the density and velocity structure simultaneously.

REFERENCES

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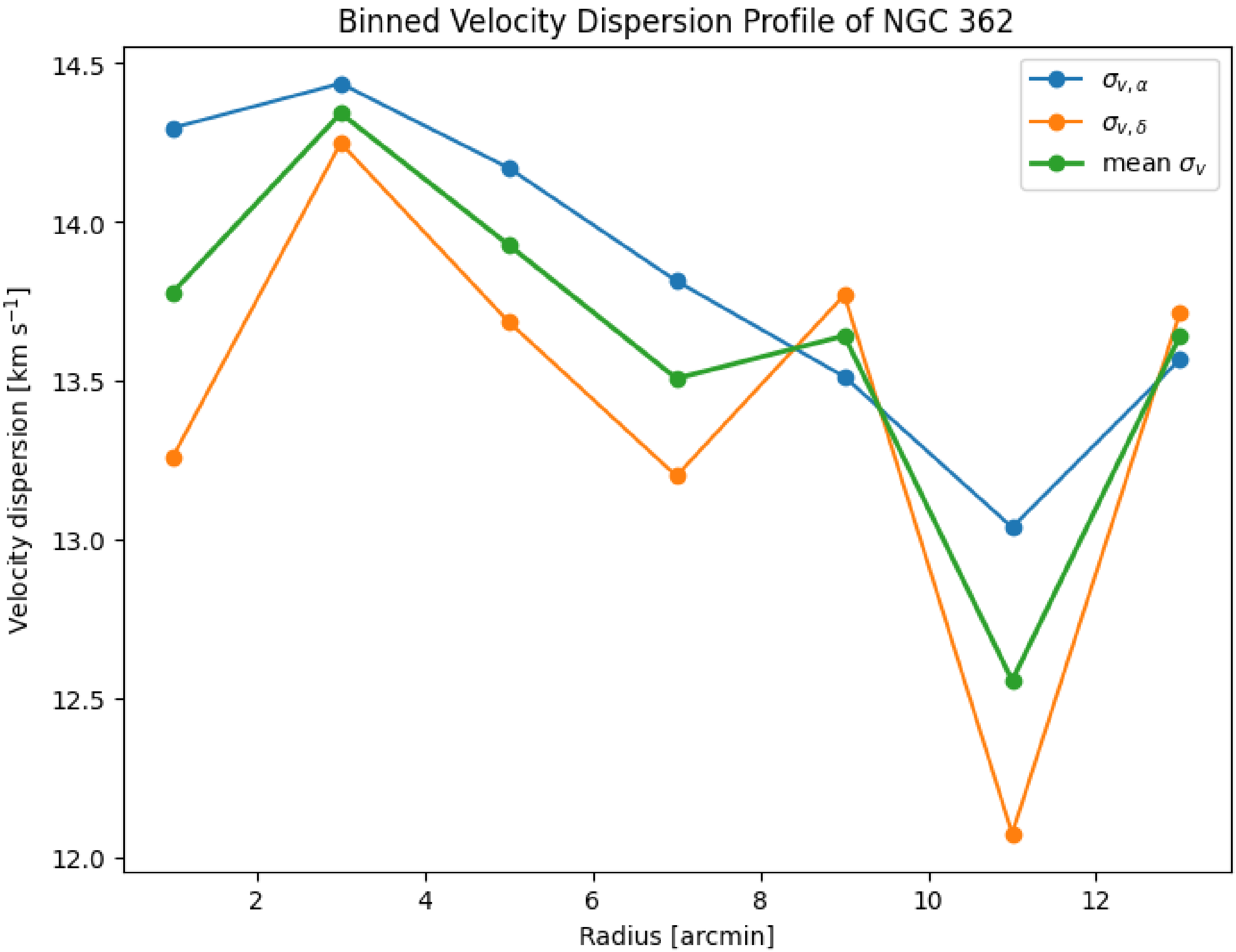

Figure 7. Binned velocity-dispersion profile of NGC 362. The proper-motion dispersions in right ascension and declination are shown separately, together with their mean in each radial bin.

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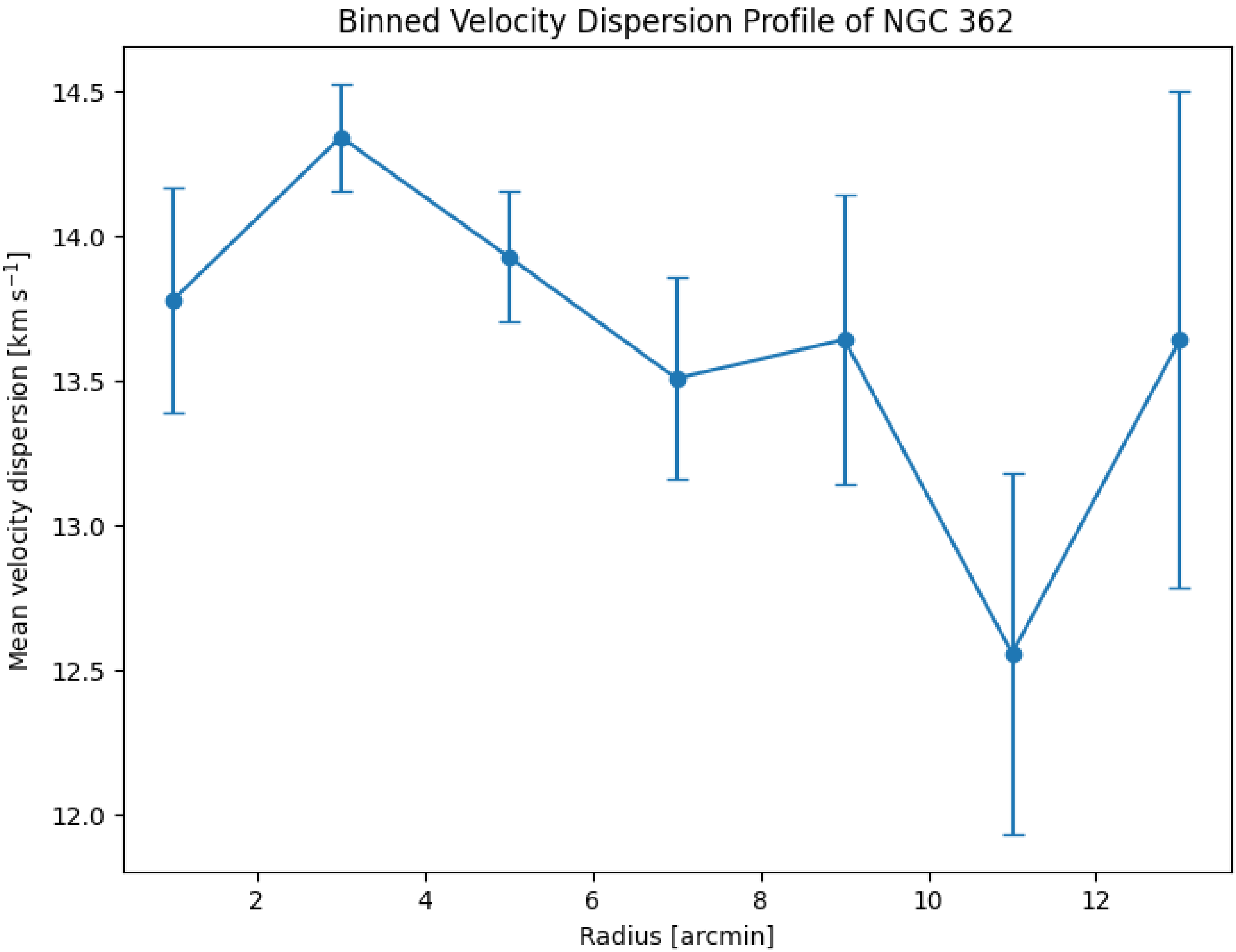

Figure 8. Mean binned velocity-dispersion profile of NGC 362 with simple statistical error bars. The profile is broadly consistent with a nearly flat dispersion, with larger uncertainties in the outer bins due to smaller numbers of stars.